

Literature Review: Sorting Boxes by Colour with a Franka Research 3 (FR3) Robot

Team: [team 06] · Members: [Youssef hazem – ID:58-2769], [imaged emil – ID:58-3830], [tony hany – ID:58-6506], [anas mohamed – ID:58-2897], [ahmed adham – ID:58-3529]

1. What we are building

We are building a robot cell that **sorts boxes by colour**. Three boxes (red, green and blue) wait on a conveyor in front of a Franka Research 3 (FR3) robot arm. The robot picks up each box with a suction cup and drops it into the bin of the same colour. The bins are on the robot's left, right and behind it, all placed inside the arm's 855 mm reach. To keep the project simple, the conveyor does not move, the box positions are read directly from the simulator (no camera), and the suction cup is simulated by attaching the box to the robot's tool while it is carried.

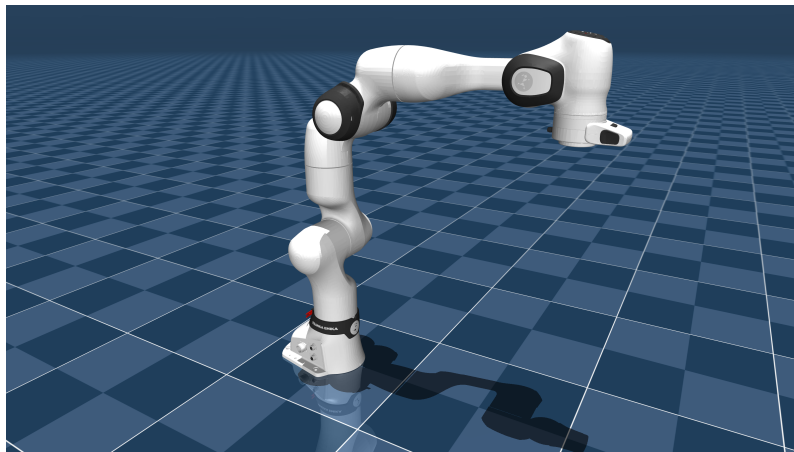


Fig. 1 – The Franka Research 3 (FR3) model from MuJoCo Menagerie [12], shown in MuJoCo. The conveyor, boxes and bins will be added around it.

2. How this problem is normally solved

Kinematics. The standard way to model a robot arm is the Denavit–Hartenberg (DH) convention: a frame is placed on each joint, and four parameters per link give the transform from one frame to the next. Multiplying these transforms gives the forward kinematics (joint angles to tool position) [1]. Inverse kinematics (tool position to joint angles) is solved either analytically with closed-form equations or numerically by iterating with the Jacobian. A 7-joint arm such as the FR3 is *redundant*: a given tool pose can be reached with infinitely many joint configurations (the elbow can swing while the tool stays still). Analytic solvers therefore fix one extra parameter, for example the angle of the last joint, and then return a finite set of solutions [2]. Numerically, the 6×7 Jacobian is inverted with the pseudo-inverse, and the leftover freedom (the null space) is used for secondary goals such as staying away from joint limits. The Jacobian also links joint velocities to tool velocities [3], [4].

Motion planning. Long moves are usually planned in joint space with cubic or quintic polynomials, so each joint starts and stops smoothly. Short moves where the path matters, like going straight down to a part, are planned as a straight line in task space and converted to joint angles with inverse kinematics [3], [4].

Control. The most common controller is a separate PD or PID loop on each joint. It is simple and works well at moderate speeds. Computed-torque control uses the robot's dynamic model to cancel gravity and inertia, so it follows the planned path more accurately [3], [4].

Simulation. The system is normally tested in a simulator before a real robot is used. Common choices are Gazebo [5], PyBullet [6] and MuJoCo [7]; MuJoCo is fast and handles contact between objects well. ROS 2 is the standard software layer that connects the control code to the robot or simulator by sending messages on named topics [8].

3. The robot we picked and why

We picked the **Franka Research 3 (FR3)**, a collaborative robot (made to work safely near people [9]) with 7 joints, a 3 kg payload and an 855 mm reach [10]. Why this robot:

1. **7 DoF**, which is inside the required 4–7 DoF range (at the upper end). The extra joint gives us a redundancy to manage, which makes the kinematics part of the project richer.
2. **Published DH parameters [10]** (in Craig's modified convention, as in [1]), so we can check our own DH table.
3. **A known closed-form inverse kinematics with one free parameter [2]** (the FR3 has the same kinematics as the Franka Emika Panda [11]), so we can check our IK results.
4. **A ready-made MuJoCo model** (`franka_fr3`) from MuJoCo Menagerie [12], so we do not have to build the model ourselves.
5. **Enough payload**: the boxes are light, well below the 3 kg limit.
6. **Reach needs care**: 855 mm is much shorter than the 1.3 m of larger arms, so the conveyor and the three bins must be placed close to the base. We will check in simulation that the arm reaches every box and every bin before we fix the layout.

4. Draft project flow

One sorting cycle (repeated until the conveyor is empty):

1. **Home**: wait in a safe pose and read the position and colour of the next box.
2. **Approach**: move to 10 cm above the box, then go straight down to it.
3. **Grasp and lift**: turn on the suction and lift the box straight up.
4. **Move**: swing smoothly to a point above the bin with the same colour.
5. **Place and return**: lower the box, turn off the suction, and go back home.

Plan: M02 – frames, DH table, forward and inverse kinematics (including how we fix the redundant joint), joint sliders in MuJoCo, start of the GUI. M03 – velocity and acceleration kinematics (Jacobian), checked against the simulator, and the full factory environment. M04 – joint-space and task-space trajectories, joint control, and the complete sorting cycle.

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